VLSI Devices
Lecture 17

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L17

Solution

- Poisson equation with boundary conditions
  - Try the following functions for the electrostatic potential.
  - For the oxide region,

$$\phi_1(x,y) = v_1(x,y) + u_{L1}(x,y) + u_{R1}(x,y)$$ Taur, Eq. (6.7)

  - For the silicon region,

$$\phi_2(x,y) = v_2(x,y) + u_{L2}(x,y) + u_{R2}(x,y)$$ Taur, Eq. (6.8)

Poisson equation with top/bottomr b.c. (Long-channel)

Laplace equation to match left b.c.

Laplace equation to match right b.c.

Solution, $v(x, y)$

- Actually, it is $v(x)$.
  - For the oxide region ($-t_{ox} \leq x \leq 0$),

$$v_1(x) = \phi_s - \frac{V_{gs} - V_{fb} - \phi_s}{t_{ox}} x$$

Taur, Eq. (6.9)

  - For the silicon region ($0 \leq x \leq W_d$),

$$v_2(x) = \phi_s \left(1 - \frac{x}{W_d}\right)^2$$

Taur, Eq. (6.10)

  - It is noted that

$$\phi_s = \frac{q N_a W_d^2}{2\epsilon_{si}}$$

Taur, Eq. (6.11)

Relation between $v_1(x)$ and $v_2(x)$

- At $x = 0$, the displacement must be continuous. (ϕ_s itself is already continuous.)
  - For the oxide region

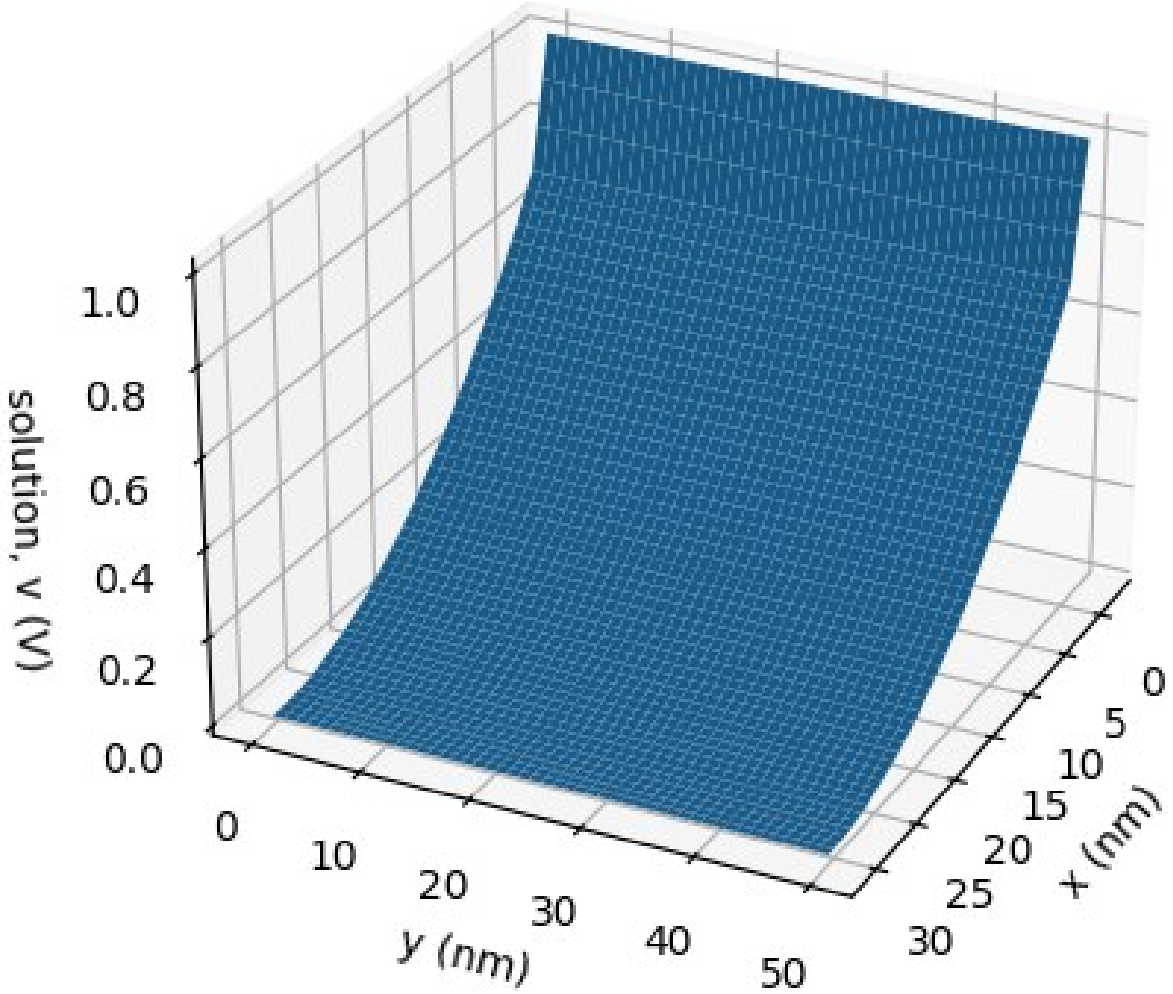
Taur, Eq. (6.12)

$$\epsilon_{ox}\frac{V_{gs} - V_{fb} - \phi_s}{t_{ox}} = \epsilon_{si}\frac{2\phi_s}{W_d} = \sqrt{2\epsilon_{si} q N_a \phi_s}$$

  - Example)
  - For our example, V_{fb} is -1.01 V, t_{ox} is 2.0 nm, N_a is 10^{18} cm^{-3}.
  - At zero gate voltage, ϕ_s is about 0.726 V.
  - The depletion width is about 30 nm.
  - (ϕ_B is about 0.474 V.)

Long-channel solution, $v_1(x)$ and $v_2(x)$

- 50-nm-long channel

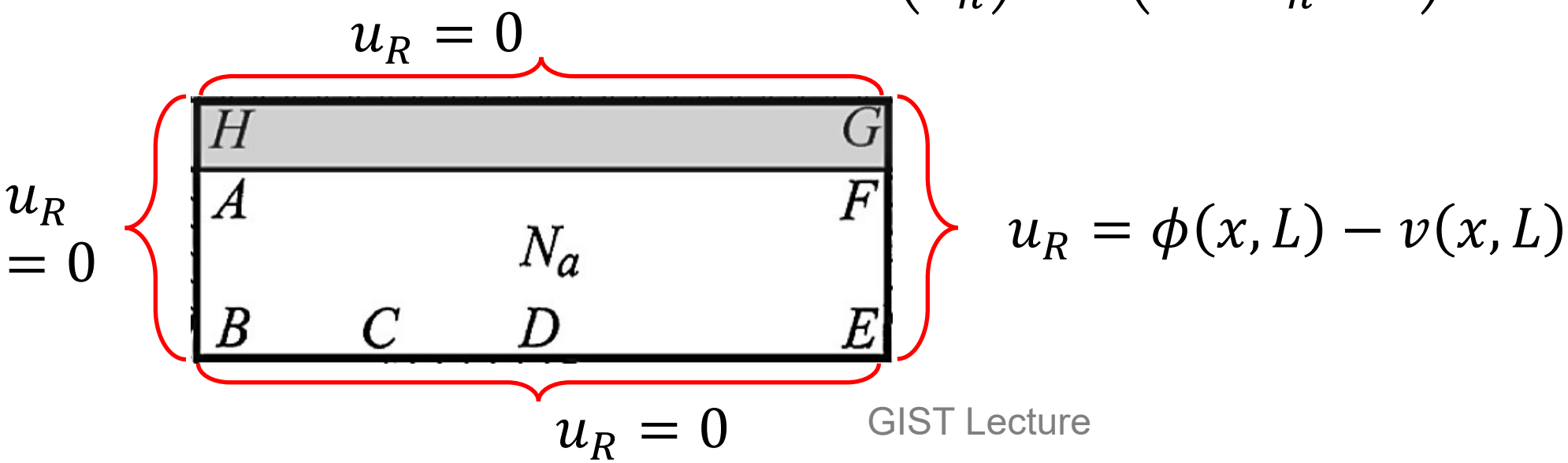


Modes for $u_{R1}(x,y)$ and $u_{R2}(x,y)$

- For three boundaries, it should vanish.

$$u_{R1,n}(x,y) = \sinh\left(\frac{\pi y}{\lambda_n}\right)\sin\left(\frac{\pi(x+t_{ox})}{\lambda_n}\right)$$ Taur, ~Eq. (6.14)

$$u_{R2,n}(x,y) = \sinh\left(\frac{\pi y}{\lambda_n}\right)\sin\left(\frac{\pi(W_d-x)}{\lambda_n}\right)$$ Taur, ~Eq. (6.16)



Series expansion of $u_R(x, y)$

- $u_R(x, L) = \phi(x, L) - v(x, L)$ can be expanded with coefficients c_n's.
  - Therefore,

$$u_{R1}(x, y) = \sum_{n=1}^{\infty} c_{1,n} \frac{\sinh\left(\frac{\pi y}{\lambda_n}\right)}{\sinh\left(\frac{\pi L}{\lambda_n}\right)} \sin\left(\frac{\pi(x + t_{ox})}{\lambda_n}\right)$$ Taur, Eq. (6.14)

$$u_{R2}(x, y) = \sum_{n=1}^{\infty} c_{2,n} \frac{\sinh\left(\frac{\pi y}{\lambda_n}\right)}{\sinh\left(\frac{\pi L}{\lambda_n}\right)} \sin\left(\frac{\pi(W_d - x)}{\lambda_n}\right)$$ Taur, Eq. (6.16)

Relation between $u_{R1}(x, y)$ and $u_{R2}(x, y)$

- At $x = 0$, the potential and the displacement must be continuous.

$$c_{1,n} \sin\left(\frac{\pi t_{ox}}{\lambda_n}\right) = c_{2,n} \sin\left(\frac{\pi W_d}{\lambda_n}\right)$$ Taur, ~Eq. (6.17)

$$c_{1,n} \epsilon_{ox} \cos\left(\frac{\pi t_{ox}}{\lambda_n}\right) = -c_{2,n} \epsilon_{si} \cos\left(\frac{\pi W_d}{\lambda_n}\right)$$ Taur, ~Eq. (6.18)

  - We don't know $c_{1,n}$, $c_{2,n}$, and λ_n.

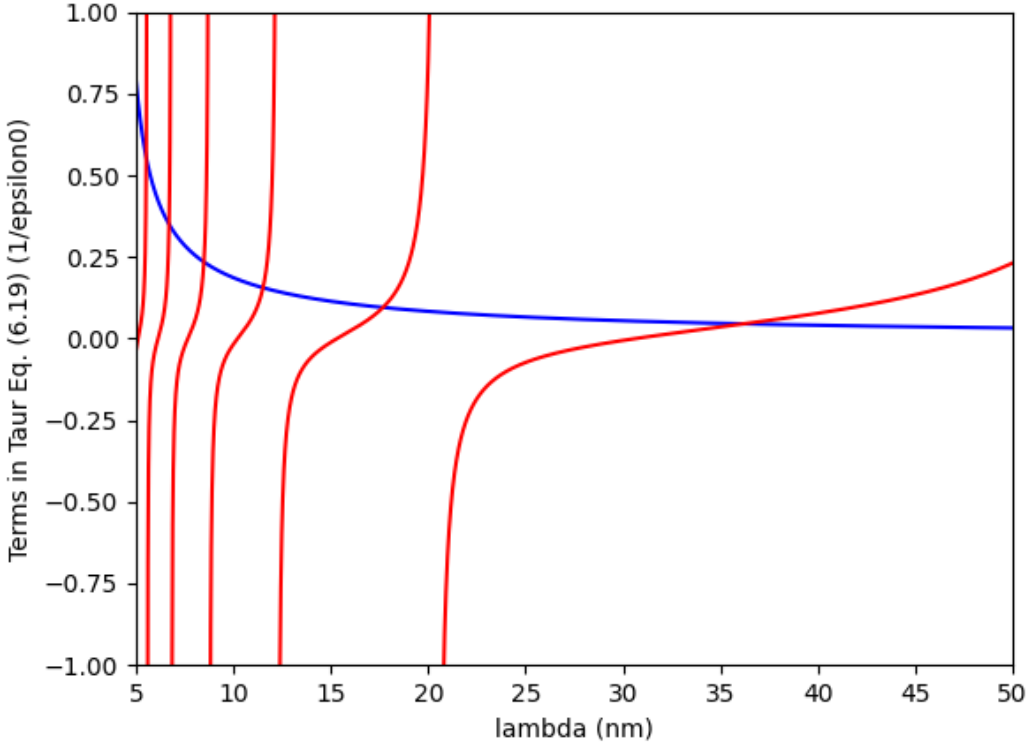
Equation for λ_n

- the potential and the displacement must be continuous.

$$\frac{1}{\epsilon_{ox}}\tan\left(\frac{\pi t_{ox}}{\lambda_n}\right) = -\frac{1}{\epsilon_{si}}\tan\left(\frac{\pi W_d}{\lambda_n}\right)$$

Taur, Eq. (6.19)

  - There are multiple solutions: $\lambda_1, \lambda_2, \ldots$
  - Example)



Homework#17

- Due: 08:00 on May 11
- Submit a PDF report through the GIST LMS system.
  - Calculate the solutions of Taur, Eq. (6.19).
  - (You may use your own parameters.)

Full 2D potential solution

- The potential and displacement must be continuous.

$$\phi_2(x,y) = \phi_s\left(1-\frac{x}{W_d}\right)^2 + \sum_{n=1}^{\infty}\frac{b_{2,n}\sinh\left(\frac{\pi(L-y)}{\lambda_n}\right)+c_{2,n}\sinh\left(\frac{\pi y}{\lambda_n}\right)}{\sinh\left(\frac{\pi L}{\lambda_n}\right)}\sin\left(\frac{\pi(W_d-x)}{\lambda_n}\right)$$

Taur, Eq. (6.20)

  - Coffecients ($b_{2,n}$ and $c_{2,n}$) are determined by the source and drain boundary conditions.

The second term of RHS

- It represents the short-channel effect.
  - Using an approximation of $\sinh(z) \approx \frac{\exp z}{2}$ for $z > 0$, the surface potential becomes

$$\frac{b_{2,n}\sinh\left(\frac{\pi(L-y)}{\lambda_n}\right) + c_{2,n}\sinh\left(\frac{\pi y}{\lambda_n}\right)}{\sinh\left(\frac{\pi L}{\lambda_n}\right)}\sin\left(\frac{\pi W_d}{\lambda_n}\right)$$

$$\approx \left[b_{2,n}\exp\left(-\frac{\pi y}{\lambda_n}\right) + c_{2,n}\exp\left(\frac{\pi(y-L)}{\lambda_n}\right)\right]\sin\left(\frac{\pi W_d}{\lambda_n}\right)$$

$$\geq 2\sqrt{b_{2,n}c_{2,n}}\exp\left(-\frac{\pi L}{2\lambda_n}\right)\sin\left(\frac{\pi W_d}{\lambda_n}\right)$$

Taur, Eq. (6.21)

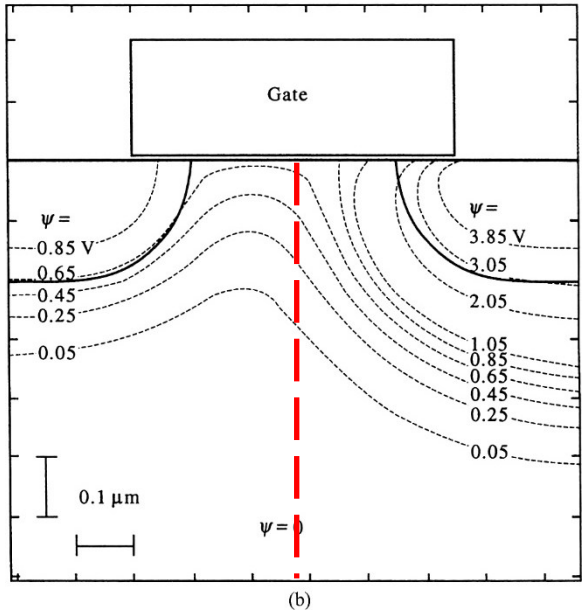
There is a point of minimum potential.

- Because of the $\exp\left(-\frac{\pi L}{2\lambda_n}\right)$ factor,
  - The longest one, λ_1, is dominant.

$$2\sqrt{b_{2,1}c_{2,1}}\exp\left(-\frac{\pi L}{2\lambda_1}\right)\sin\left(\frac{\pi W_d}{\lambda_1}\right)$$

  - This minimum point is found at

$$y_{min} = \frac{L}{2} - \frac{\lambda_1}{2\pi}\ln\left(\frac{c_{2,1}}{b_{2,1}}\right)$$



Potential profile
(Taur, Fig. 6.2(b))

Minimum channel length

- When approximated forms of $b_{2,1}$ and $c_{2,1}$ are used,
  - The minimum surface potential is found as

$$\phi_2(0, y_{min}) \approx \phi_s + \frac{8}{\pi}\sqrt{(\phi_{bi} - 0.5\phi_s)(\phi_{bi} + V_{ds} - 0.5\phi_s)}\sin\left(\frac{\pi W_{dm}}{\lambda_1}\right)\exp\left(-\frac{\pi L}{2\lambda_1}\right)$$

Taur, Eq. (6.35)

  - Then, the minimum channel length is $L_{min} \approx 2\lambda_1$.

Scale length, $\lambda = \lambda_1$

- Since λ_1 is dominant, it is defined as the scale length.
  - The scale length equation:

$$\frac{1}{\epsilon_{ox}} \tan\left(\frac{\pi t_{ox}}{\lambda}\right) + \frac{1}{\epsilon_{si}} \tan\left(\frac{\pi W_{dm}}{\lambda}\right) = 0$$

Taur, Eq. (6.23)

  - λ is larger than both of W_{dm} and t_{ox}.
  - When $t_{ox} \ll W_{dm}$, $\lambda \approx W_{dm} + \frac{\epsilon_{si}}{\epsilon_{ox}} t_{ox}$.

Thank you!